PCRNet: Progressive Correlation Refinement for Few-Shot Hyperspectral Image Classification

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***Abstract*—Few-shot hyperspectral image classification (FHSIC) is crucial for precise land cover identification when labeled samples are scarce, mitigating the prohibitive annotation costs associated with hyperspectral image (HSI) data. However, existing FHSIC methods primarily follow a feature refinement paradigm, which struggles to effectively model the holistic relational structure among samples. Meanwhile, the commonly used attention mechanism tends to focus solely on sample-specific local information, limiting its ability to capture global category knowledge and promote cross-domain generalization. To address these challenges, we propose an efficient progressive correlation refinement network (PCRNet). Our work pioneers a correlation refinement (CR) paradigm, shifting the objective from enhancing feature representations to explicitly reasoning upon the structured query–class relationships. Specifically, PCRNet first constructs a learnable generalized correlation matrix that encodes the intersample relationships. Then, we present a simple yet effective memory-guided global attention (MGA) mechanism to facilitate the injection of category commonality priors into the relational structure from a holistic perspective. This is further combined with query-level adaptive modulation (QAM) and symmetrical gated integration (SGI) modules to achieve progressive refinement of the generalized correlation matrix. Finally, the refined correlation is projected as the classification logit, enabling accurate identification of land covers. Experiments on four benchmark HSI datasets demonstrate the state-of-the-art performance of our approach. We believe that this new paradigm opens a promising direction for future FHSIC model design. Code is available at https://github.com/guoying918/PCRNet**

***Index Terms*—Attention mechanism, correlation modeling, few-shot learning (FSL), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs), acquired by imaging spectrometers, are rich 3-D data cubes encompassing hundreds or even thousands of contiguous spectral bands. These images uniquely synergize detailed spectral signatures with intricate spatial contextual information, enabling extensive applications in diverse fields such as target detection [1], mineral exploration [2], and environmental monitoring [3]. A fundamental task within HSI analysis is HSI classification (HSIC), which aims to assign precise categorical labels to ground objects by effectively leveraging the spectral characteristics and spatial contextual features inherent in these 3D data cubes. In recent years, HSIC has attracted a surge of research attention [4], [5], [6], [7].

The inherent characteristics of HSI data, namely, high dimensionality and strong interband redundancy, pose significant challenges for traditional classification methods, e.g., support vector machine [15]. These methods often struggle to effectively exploit the rich spatial–spectral information embedded within the HSI data, leading to performance bottlenecks such as the Hughes phenomenon [16]. The advent and rapid advancement of deep learning technology, with its inherent advantages in automatic feature learning and end-to-end training, have brought breakthroughs to HSIC. Following the initial success of stacked autoencoders [17] in HSI feature extraction, a diverse array of deep network architectures, including convolutional neural networks (CNNs) [18], [19], [20], recurrent neural networks (RNNs) [21], and graph convolutional networks (GCNs) [22], [23], has been widely applied in extracting salient spatial–spectral features. Recently, the integration of

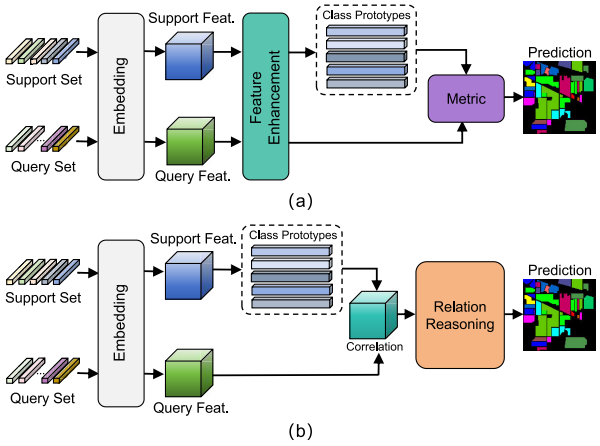


Fig. 1. Comparisons between previous feature refinement and our correlation refinement (CR) paradigms. (a) Prevailing FHSIC methods [8], [9], [10], [11], [12], [13], [14] assign class labels by applying an explicit distance metric to features that have been extracted and subsequently enhanced, essentially optimizing the feature representations of support or query samples. (b) We propose a novel CR paradigm that models a learnable generalized correlation matrix between support and query samples and then progressively optimizes it, thereby achieving a deeper reasoning of their interaction relationships.

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advanced sequence modeling approaches—particularly attention mechanisms [24], [25], [26], [27], [28] as exemplified in Transformers [11], [29], [30], [31], alongside emerging architectures such as Mamba [32], [33]—has further enhanced feature representation and boosted overall HSIC performance. Although these deep models perform well when trained with abundant labeled data, their generalizability is limited in scenarios with scarce labeled samples. Identifying new categories often necessitates collecting substantial HSI data and performing meticulous manual labeling, which is not only time-consuming and labor-intensive but also severely curtails the scalability and adaptability of HSIC in practical applications. This critical challenge underscores the need for effective few-shot HSIC (FHSIC) solutions.

To alleviate the challenge of scarce labeled samples, early attempts include data augmentation techniques [34], [35], [36] and the development of low-parameter models [37], [38]. However, these approaches suffer from suboptimal classification accuracies and restricted generalization capabilities, primarily due to their reliance on limited prior knowledge about the data distribution and the insufficient expressive capacity of the models. In recent years, few-shot learning (FSL) has emerged as a powerful and widely adopted strategy to mitigate the burden of data collection and annotation. Unlike supervised learning, FSL typically employs a metalearning framework, training models by constructing a series of metatasks where support and query sets serve as basic units. The entire process includes metatraining and metatesting stages to fully exploit prior knowledge about metatasks. By training on a diverse set of metatasks, the model acquires the ability to "learn to learn," enabling rapid adaptation to new categories with few labeled examples. Among various FSL approaches, metric-based metalearning models [39], [40], [41], [42] have shown particularly encouraging performance in FHSIC. These methods generally stack a hierarchy of deep network modules to project input samples into a common embedding space and then classify query samples by measuring their distance (i.e., feature similarity) to support samples.

Despite the promising performance of existing metric-based FHSIC methods, they still face two obvious issues. **1)** Current methods [8], [9], [10], [11], [12], [13], [14] are conceptually bound by a feature refinement paradigm. The key process is to learn highly discriminative feature representations for individual support or query samples (e.g., by incorporating sophisticated feature enhancement modules), which are then fed into a classifier, as shown in Fig. 1(a). While methods inspired by relation networks (RNs) [43], [44], [45], [46] replace fixed distance metrics (e.g., Euclidean or Mahalanobis distance [8], [9]) with a learnable module, they still operate on isolated feature pairs, performing a one-step mapping from features to a scalar relation score. This "representation-metric" approach inherently limits their ability to reason about the holistic relational structure among samples. Therefore, existing methods focus on generating more discriminative feature representations for individual samples, thereby overlooking the rich contextual information that exists across the entire set of query samples and global class knowledge. **2)** Existing methods often employ attention mechanisms to model relationships, for instance, by applying self-attention within individual HSI samples to emphasize salient spatial–spectral features [10], [13], or by using cross-attention to model category relationships between specific support and query samples [9], [11], [14]. However, such attention modeling typically operates at the individual sample level, focusing on local information pertinent to that specific sample. This localized perspective inevitably limits their ability to capture and utilize broader, global category commonality information and the overall feature distribution across the entire dataset, thereby impeding robust cross-domain knowledge transfer.

To overcome these limitations, we depart from the conventional feature refinement paradigm and introduce a novel correlation refinement (CR) paradigm, as depicted in Fig. 1(b). Our core design philosophy is to elevate the generalized correlation matrix (i.e., a structured representation of all query–class relationships) to be the primary object of learning and reasoning. Our proposed progressive CR network (PCRNet) first constructs an initial generalized correlation matrix as a preliminary estimation. Then, instead of a single mapping, it employs a series of residual CR group (RCRG) blocks to iteratively refine this matrix by integrating multilevel contextual cues. Specifically, each RCRG incorporates several key components. *First*, we propose a tractable memory-guided global attention (MGA) mechanism. MGA constructs a unified and parameterized class-specific global memory unit (CGMU) designed to learn and store discriminative categorical information that encapsulates common characteristics from the entire source or target domain (TD) dataset, effectively forming a standardized class-specific global dictionary. By computing classwise attention between an input sample and the CGMU, the class-level global augmentation (CGA) is achieved, enabling the model to leverage global category knowledge from a holistic, dataset-wide perspective. *Second*, a query-level adaptive modulation (QAM) module is developed, which utilizes a standard self-attention mechanism to establish fine-grained relationships among query samples. This allows the model to enforce relational consistency across different queries within the correlation matrix. *Third*, information from both the class level and the query level is synergistically fused using a symmetrical gated integration (SGI) module. This integrated approach facilitates the progressive optimization of the generalized correlation matrix, leading to a more globally coherent relational structure and, thus, improving the accuracy of FHSIC. Extensive experiments on four benchmark datasets demonstrate that our PCRNet significantly outperforms existing state-of-the-art methods, validating the superiority of the proposed paradigm shift.

In summary, our contributions are given as follows.

1) We identify and formulate a novel CR paradigm for FHSIC for the first time, shifting the focus from feature learning to explicit relational reasoning. This provides a new perspective for tackling the FHSIC task.
2) We design an effective RCRG block, which acts as the core engine of our paradigm, synergistically integrating global class priors (via CGA) and interquery context (via QAM) to enable progressive optimization of the correlation structure.

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3) We propose a tractable MGA mechanism that adaptively learns a stable, dataset-wide global memory unit, providing robust categorical priors that enhance cross-domain generalization.

The rest of this article is organized as follows. Section II briefly introduces the related work of FHSIC. Section III describes the proposed method in detail. Experimental details and results on benchmark datasets are presented in Section IV. The distinction from traditional relational networks is subsequently discussed in Section V. Finally, a conclusion is drawn in Section VI.

II. Related Work

The challenge of classifying HSIs with limited labeled samples has spurred significant research efforts. This section reviews the relevant literature from two key perspectives: FHSIC and the feature refinement paradigm in FHSIC.

A. FHSIC

FHSIC aims to enable models to generalize effectively to novel classes from a paucity of labeled examples. Among various FHSIC approaches, metalearning has emerged as a cornerstone. Within the metalearning landscape, one major branch consists of optimization-based methods, such as model-agnostic metalearning [47]. These methods seek to learn an optimal parameter initialization that allows for rapid adaptation to new tasks via a few gradient updates. Another dominant branch is composed of metric-based methods, which focus on learning a discriminative embedding space for similarity-based classification, as detailed in Section II-B. Beyond metalearning, other notable strategies for FHSIC include transfer learning with fine-tuning [48], [49] although its performance is often constrained by the domain shift between source and target datasets [50]. Data augmentation techniques, such as pixel-pair strategy [51] and GAN-based sample synthesis [52], are also widely investigated to address data scarcity. To effectively exploit the rich spatial–spectral information inherent in HSIs, these diverse FHSIC methods are commonly built upon advanced network backbones, including CNNs for local feature extraction [8], [12], GCNs for modeling contextual relationships [53], [54], and, more recently, Transformers for capturing long-range dependencies [10], [11], [55].

B. Feature Refinement Paradigm in FHSIC

Metric learning has become a dominant approach in FHSIC, aiming to learn a feature space where samples from the same class are clustered compactly, while those from different classes are well-separated [56]. Prevailing frameworks, frequently adapted for FHSIC, include Siamese networks [39], matching networks [40], prototypical networks [41], and RNs [42]. These methods learn to classify query samples by measuring similarity to support samples within the learned embedding space, either via fixed distance metrics or a learned, nonlinear metric embodied by a learnable deep network. In particular, RNs [46] and their variants in FHSIC [44], [45] represent a significant advancement, replacing fixed distance metrics with a learnable deep network. However, it remains within the feature refinement paradigm as its fundamental operation is still a direct, one-pass mapping from isolated feature pairs to a scalar relation score, without explicitly modeling or refining the global relational context. In the context of FHSIC, many studies have adapted these frameworks by designing specialized spatial–spectral embedding modules. For instance, researchers have used attention mechanisms to refine class prototypes [22], developed convolutional RNs to establish relationships between feature maps and categories [43], and explored distillation-constrained learning to enhance discriminability [57]. Other efforts focus on more sophisticated metric loss functions or transductive inference mechanisms to further boost performance [11], [58], [59]. Despite these advancements, a common thread in metric-based methods is their implicit reliance on a feature refinement paradigm, where the primary goal is to enhance the quality of individual sample representations. This focus overlooks the potential of treating the entire landscape of sample relationships as a structured object for explicit modeling and reasoning. In contrast, our proposed approach is designed to address this gap by developing an effective CR paradigm.

III. Methodology

In this section, we first provide an overview of the overall framework of our proposed PCRNet in Section III-A. Then, we employ a lightweight embedding network in Section III-B to project support and query samples into a discriminative feature space. Subsequently, a generalized correlation matrix is constructed in Section III-C to represent the interaction relationship between query samples and support class prototypes. This correlation is then progressively refined by cascading three RCRG blocks, as detailed in Section III-D. Note that each RCRG block comprises three key modules: CGA, QAM, and SGI. Finally, the correlation projection is made in Section III-E for the final classification of the land cover.

A. Overall Framework

Unlike conventional RNs that perform a one-step mapping from isolated feature pairs to a scalar relation score, our PCRNet elevates the entire structured correlation matrix between all query–class pairs as the primary object of learning and refinement. This enables holistic relational reasoning beyond pairwise comparisons, capturing both global class commonalities and interquery dependencies.

Our PCRNet operates within the standard episode-based metalearning framework, as shown in Fig. 2. In the workflow diagram, we annotate the major stages using numbered step markers (①–③), where ① denotes data processing, ② denotes meta-training, and ③ denotes meta-testing. In the stage ①, we follow the metalearning data processing strategy. Each episode is partitioned into a support set $\mathcal{S}$ and a query set $\mathcal{Q}$ to mimic a C-way K-shot metatask, where C denotes the number of classes and K represents the number of labeled samples per class. Specifically, following [8], [9], and [11], we perform data augmentation on the novel class samples in the TD by adding random Gaussian noise. Subsequently,

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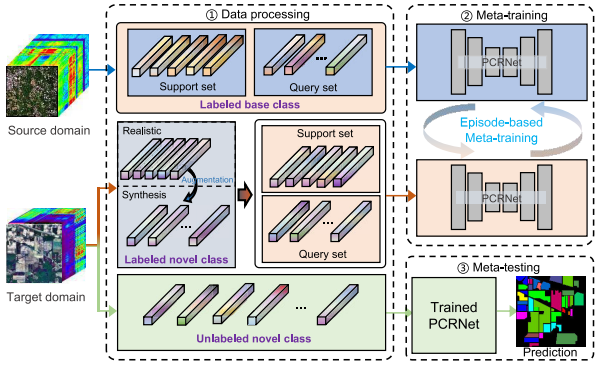


Fig. 2. Workflow of our proposed method for FHSIC, built upon the episode-based metalearning framework. For clarity, the one-way five-shot setting is presented.

K samples from each class are randomly selected to form the support set $\mathcal{S} = \{\mathbf{X}_i\}_{i=1}^{N_s}$, while the remaining N samples per class constitute the query set $\mathcal{Q} = \{\mathbf{X}_i\}_{i=N_s+1}^{N_s+N_q}$. Here, $\mathbf{X}_i \in \mathbb{R}^{P\times P\times B}$ is the ith sample, represented as an image patch, where $P \times P$ denotes the spatial window size and B is the number of spectral bands. As such, $\mathcal{S}$ contains a total of $N_s = C \times K$ samples, and $\mathcal{Q}$ contains $N_q = C \times N$ samples. Then, we advocate the transfer of metaknowledge from base classes with abundant annotations to novel classes with limited annotations through episode-based collaborative metatraining between disjoint source and TD datasets in the stage ②. During the stage ③, unlabeled novel class samples are directly fed into the trained PCRNet to predict the pixelwise classification map.

The core innovation of PCRNet lies in the introduction of a novel CR paradigm, which shifts the network's learning focus from conventional feature representation learning to the explicit modeling and optimization of interaction relationships between support and query samples. As illustrated in Fig. 3(a), the pipeline of PCRNet is given as follows. First, we utilize an embedding network to extract feature representations from support and query samples, respectively. Based on these embeddings, a structured, learnable generalized correlation matrix is constructed to characterize the initial interaction relationships between query features and support class prototypes. Next, this initial correlation matrix is fed into a cascade of RCRG blocks for coarse-to-fine refinement. Each RCRG block comprises two parallel branches (i.e., CGA and QAM) and an integration module (i.e., SGI), working synergistically to enhance the correlation representation.

1) *CGA:* As shown in Fig. 3(b), this branch aims to inject category commonality knowledge from a holistic perspective. Its core is our proposed MGA mechanism. MGA learns and stores category-level information that reflects the commonalities across the entire dataset by constructing a parameterized CGMU, which functions as a global dictionary. CGA leverages this dictionary to provide a cross-domain robust, global category prior for the optimization of classwise correlations.
2) *QAM:* Operating in parallel with CGA, the QAM module in Fig. 3(c) leverages self-attention mechanisms to focus on modeling fine-grained dependencies within the query samples. It adaptively captures the intersample context, thereby refining querywise correlations.
3) *SGI:* Following this, SGI employs a gating strategy to symmetrically and adaptively fuse the correlation clues from both CGA and QAM modules, as illustrated in Fig. 3(d). This facilitates a synergy between classwise and querywise information.

Finally, the refined correlation representation is used to guide the final classification decision through a correlation projection module, enabling the learning of a more optimal decision boundary in few-shot conditions and boosting the model's generalization capability.

B. Feature Embedding

The inherent high-dimensionality of HSI data leads to significant spectral redundancy, which degrades feature utility and increases the computational burden. To enhance feature compactness and discriminability, we introduce an embedding network to map the HSI data into a low-dimensional feature space. In practice, to strike a balance between feature representation and computational efficiency, this network typically avoids processing the entire HSI image and, instead, captures spatial contexts by extracting local image patches. Specifically, for each central pixel (x_i, y_i) targeted for classification, an image patch $\mathbf{X}_i \in \mathbb{R}^{P\times P\times B}$ centered at (x_i, y_i) is leveraged. $\mathbf{X}_i$ serves as the input to the feature embedding module, enabling the capture of representative spatial–spectral features.

We employ a lightweight MICD backbone proposed in [12] as our feature embedding module $\mathcal{F}$. Notably, $\mathcal{F}$ is replaceable (see Section IV-E), which ensures the flexibility and scalability of our PCRNet. We project the support samples $\mathbf{X}^{\mathcal{S}} \in \mathbb{R}^{N_s\times P\times P\times B}$ and query samples $\mathbf{X}^{\mathcal{Q}} \in \mathbb{R}^{N_q\times P\times P\times B}$ into a shared D-dimensional feature space, yielding the support features $\mathbf{F}^{\mathcal{S}} \in \mathbb{R}^{N_s\times D}$ and query features $\mathbf{F}^{\mathcal{Q}} \in \mathbb{R}^{N_q\times D}$, respectively. Formally,

$$\mathbf{F}^{\mathcal{S}} = \left\{\mathcal{F}\left(\mathbf{X}_i^{\mathcal{S}}\right)\right\}_{i=1}^{N_s}, \quad \mathbf{F}^{\mathcal{Q}} = \left\{\mathcal{F}\left(\mathbf{X}_i^{\mathcal{Q}}\right)\right\}_{i=1}^{N_q}. \tag{1}$$

Therefore, the support class prototypes $\mathbf{P}^{\mathcal{S}} \in \mathbb{R}^{N_c\times D}$ can be obtained by averaging the K support features associated with each class, where $N_c \triangleq C$ denotes the number of classes for subsequent representation.

C. Generalized Correlation Matrix Construction

As the cornerstone of our CR paradigm, we first construct an initial generalized correlation matrix with a structured form. This matrix is designed to explicitly encode the interaction relationships between query samples and support class prototypes, allowing its parameters to be updated through a deep network. To this end, we directly combine the query features $\mathbf{F}^{\mathcal{Q}} \in \mathbb{R}^{N_q\times D}$ and the support class prototypes $\mathbf{P}^{\mathcal{S}} \in \mathbb{R}^{N_c\times D}$, as well as computing their explicit distance. Concretely, $\mathbf{F}^{\mathcal{Q}}$ is expanded to $\tilde{\mathbf{F}}^{\mathcal{Q}} \in \mathbb{R}^{N_q\times N_c\times D}$ to align each query feature with all support class prototypes. Similarly, $\mathbf{P}^{\mathcal{S}}$ is also expanded to $\tilde{\mathbf{P}}^{\mathcal{S}} \in \mathbb{R}^{N_q\times N_c\times D}$ to match the expanded query features. In particular, we additionally introduce an explicit distance tensor $\mathcal{D}(\tilde{\mathbf{F}}^{\mathcal{Q}}, \tilde{\mathbf{P}}^{\mathcal{S}}) \in \mathbb{R}^{N_q\times N_c\times 1}$ to guide the initial correlation modeling, which is computed by the Mahalanobis distance

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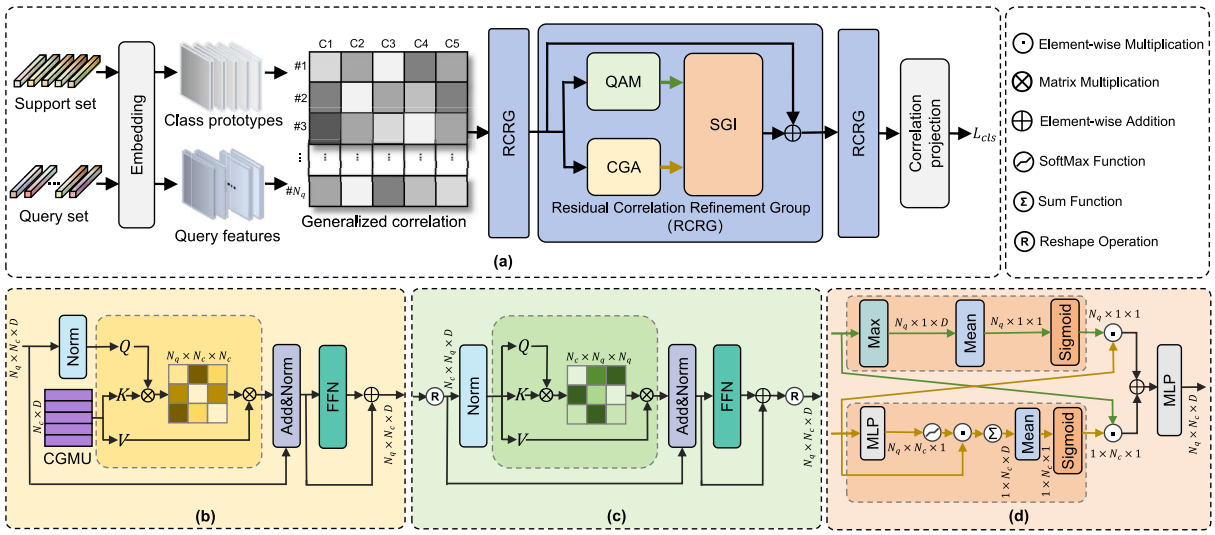


Fig. 3. Overall network architecture. (a) Our PCRNet consists of three key processing steps. First, an embedding network extracts features from both support and query samples. Then, an initial generalized correlation matrix is constructed, capturing the interaction relationship between query features and support class prototypes. Finally, the RCRG block is employed to progressively optimize this correlation matrix, followed by a correlation projection module to achieve land cover classification. The entire model is supervised using a cross-entropy loss. Within the RCRG, (b) CGA leverages a parameterized CGMU as a class-specific global dictionary to enhance classwise correlations from a holistic perspective, and (c) QAM is designed to refine querywise correlations via self-attention. The refined correlations from these two branches are then adaptively aggregated by (d) SGI. This figure illustrates a five-way one-shot setting.

used in [8], [9], and [14]. Next, these three components are concatenated along the channel dimension to produce an initial generalized correlation matrix with a dimension of $N_q \times N_c \times (2D+1)$. Finally, we apply an MLP, composed of two linear layers and a ReLU activation, to transform this initial matrix into a D-dimensional correlation representation, yielding

$$\mathbf{M}_{\mathrm{GCM}} = \mathrm{MLP}\left(\tilde{\mathbf{F}}^{\mathcal{Q}} \,\|\, \tilde{\mathbf{P}}^{\mathcal{S}} \,\|\, \mathcal{D}\left(\tilde{\mathbf{F}}^{\mathcal{Q}}, \tilde{\mathbf{P}}^{\mathcal{S}}\right)\right) \tag{2}$$

where $\|$ is a concatenation operation in the channel dimension. In this manner, we construct a structured and learnable generalized correlation matrix, denoted as $\mathbf{M}_{\mathrm{GCM}} \in \mathbb{R}^{N_q \times N_c \times D}$. This initial matrix contains only pairwise, local information; the goal of the subsequent refinement process is to enrich it with global and contextual knowledge.

D. Residual CR Group

To further enhance the captured interaction relationships, the initial generalized correlation matrix, denoted as $\mathbf{M}^0_{\mathrm{GCM}}$, is progressively optimized by a stack of L cascaded RCRG blocks. Within each RCRG block, the correlation matrix from the previous stage, $\mathbf{M}^{l-1}_{\mathrm{GCM}}$, is processed by two parallel branches: CGA and QAM. Notably, the CGA module, underpinned by the MGA mechanism, injects global class commonalities from a holistic, dataset-wide perspective. Concurrently, the QAM module focuses on enforcing relational consistency among queries. Subsequently, the SGI module synergistically fuses the complementary correlation clues from these two branches. This process for RCRG can be expressed as follows:

$$\tilde{\mathbf{M}}^{l}_{\mathrm{GCM}} = \mathrm{SGI}\left(\mathrm{QAM}\left(\mathbf{M}^{l-1}_{\mathrm{GCM}}\right), \mathrm{CGA}\left(\mathbf{M}^{l-1}_{\mathrm{GCM}}, \mathbf{M}_{\mathrm{CGMU}}\right)\right) \tag{3}$$

where $\mathbf{M}_{\mathrm{CGMU}} \in \mathbb{R}^{N_c \times D}$ denotes a CGMU, which serves as a global dictionary to learn and store dataset-specific class commonality information. Note that due to the disjoint classes between the source and TD datasets, the parameters of $\mathbf{M}_{\mathrm{CGMU}}$ are not shared between them. To mitigate information degradation during iterative refinement and facilitate robust training, a residual connection is employed within each RCRG block, producing the refined output $\mathbf{M}^{l}_{\mathrm{GCM}}$, i.e.,

$$\mathbf{M}^{l}_{\mathrm{GCM}} = \mathbf{M}^{l-1}_{\mathrm{GCM}} + \tilde{\mathbf{M}}^{l}_{\mathrm{GCM}}. \tag{4}$$

The final output after L RCRG blocks is $\mathbf{M}^{L}_{\mathrm{CGMU}}$. To balance efficiency and accuracy, we empirically set L to 3, as discussed in Section IV-D. In the following, we detail these three key modules in the RCRG block and omit the superscript $l-1$ for simplicity.

1) Class-Level Global Augmentation: The CGA module, powered by the proposed MGA mechanism, aims to enrich the input $\mathbf{M}^{l-1}_{\mathrm{GCM}} \in \mathbb{R}^{N_q \times N_c \times D}$ by injecting dataset-wide, global class priors. Notably, the MGA mechanism enhances cross-domain generalization by providing a domain-invariant anchor. Specifically, MGA leverages a learnable, parameterized CGMU, $\mathbf{M}_{\mathrm{CGMU}} \in \mathbb{R}^{N_c \times D}$, initialized randomly. It is jointly optimized with the network via standard backpropagation during the metatraining phase. The ability of CGMU to capture domain-invariant knowledge arises from the shared spectral–physical characteristics of land covers across HSI datasets. Although spatial resolutions or specific environments change between domains, the fundamental spectral relationships (e.g., the relative correlation patterns between "vegetation" and "soil") remain structurally similar, as detailed in Section III of the supplementary material. Different from class prototypes (i.e., local averages of support features) used in prototypical networks, the CGMU is a dataset-wide memory unit. It acts as a learnable global class dictionary, tasked with distilling and storing discriminative class commonalities from the entire source or TD dataset, as demonstrated in Section I

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of the supplementary material. Functionally, instead of serving as a direct metric anchor for classification, the CGMU acts as a stable global prior to guide the refinement of the relational structure, which is particularly crucial for enhancing network modeling capabilities under limited-sample conditions. By computing classwise attention between the input generalized correlation matrix and this global dictionary, CGA enriches the correlation representation, infusing the refinement process with robust and holistic class knowledge.

Specifically, as shown in Fig. 3(b), the input $\mathbf{M}_{\mathrm{GCM}}$ is linearly projected to form the *query* $\mathbf{Q}_c \in \mathbb{R}^{N_q \times N_c \times D'}$. Meanwhile, the global CGMU $\mathbf{M}_{\mathrm{CGMU}}$ is linearly projected and expanded to generate the *key* $\mathbf{K}_c \in \mathbb{R}^{N_q \times N_c \times D'}$ and *value* $\mathbf{V}_c \in \mathbb{R}^{N_q \times N_c \times D'}$. Note that $D' = D/N_h$, where N_h is the number of heads, which we set to four empirically. Formally,

$$\mathbf{Q}_c = \mathbf{M}_{\mathrm{GCM}}\mathbf{W}_c^Q, \quad \mathbf{K}_c = \mathbf{M}_{\mathrm{CGMU}}\mathbf{W}_c^K, \ \mathbf{V}_c = \mathbf{M}_{\mathrm{CGMU}}\mathbf{W}_c^V \tag{5}$$

where $\mathbf{W}_c^Q, \mathbf{W}_c^K, \mathbf{W}_c^V \in \mathbb{R}^{D \times D'}$ are learnable projection matrices. Next, the resulting attention output is then processed by a feed-forward network (FFN) to generate the refined correlation matrix, which can be formulated as follows:

$$\mathbf{M}_{\mathrm{CGA}} = \mathrm{FFN}\left(\mathrm{Softmax}\left(\frac{\mathbf{Q}_c\mathbf{K}_c^\top}{\gamma_c}\right)\mathbf{V}_c\right) \tag{6}$$

where $\top$ is the transpose operation and γ_c denotes a scaling parameter to control the magnitude of the dot product. In line with the standard multihead attention architecture, this process incorporates an output merging, layer normalization, and a residual connection around the attention and FFN blocks, yielding the CGA-refined correlation matrix $\mathbf{M}_{\mathrm{CGA}} \in \mathbb{R}^{N_q \times N_c \times D}$. In this way, our MGA mechanism enables the CGA module to imbue the correlation matrix with robust global priors, thereby enhancing the cross-domain knowledge transfer capabilities.

2) Query-Level Adaptive Modulation: The QAM module is designed to refine the input $\mathbf{M}_{\mathrm{GCM}}^{l-1} \in \mathbb{R}^{N_q \times N_c \times D}$ from a complementary perspective: modeling fine-grained, intraquery dependencies. The rationale is that the correlation of a specific query sample with various class prototypes can be calibrated by observing how other query samples relate to those same prototypes. To this end, this module employs a multihead self-attention mechanism operating across the query dimension.

Specifically, as shown in Fig. 3(c), the input $\mathbf{M}_{\mathrm{GCM}}$ is first transposed to $\tilde{\mathbf{M}}_{\mathrm{GCM}} \in \mathbb{R}^{N_c \times N_q \times D}$. This rearranges the data to treat, for each class, the set of N_q query correlation vectors as a sequence upon which attention can be performed. Subsequently, $\tilde{\mathbf{M}}_{\mathrm{GCM}}$ is linearly projected to generate the *query* $\mathbf{Q}_q$, the *key* $\mathbf{K}_q$, and the *value* $\mathbf{V}_q$, all of which have a dimension of $N_c \times N_q \times D'$. The formula is given as follows:

$$\mathbf{Q}_q = \tilde{\mathbf{M}}_{\mathrm{GCM}}\mathbf{W}_q^Q, \quad \mathbf{K}_q = \tilde{\mathbf{M}}_{\mathrm{GCM}}\mathbf{W}_q^K, \ \mathbf{V}_q = \tilde{\mathbf{M}}_{\mathrm{GCM}}\mathbf{W}_q^V \tag{7}$$

where $\mathbf{W}_q^Q, \mathbf{W}_q^K, \mathbf{W}_q^V \in \mathbb{R}^{D \times D'}$ are learnable projection matrices. Then, the self-attention can be similarly calculated as

$$\mathbf{M}_{\mathrm{QAM}} = \mathrm{FFN}\left(\mathrm{Softmax}\left(\frac{\mathbf{Q}_q\mathbf{K}_q^\top}{\gamma_q}\right)\mathbf{V}_q\right) \tag{8}$$

where γ_q is a scaling factor. Finally, the resulting tensor is transposed back to its original shape of $N_q \times N_c \times D$ to yield the QAM-refined correlation matrix $\mathbf{M}_{\mathrm{QAM}} \in \mathbb{R}^{N_q \times N_c \times D}$. As with CGA, we adopt output merging, layer normalization, and residual connections. This process allows QAM to dynamically model interquery relationships at the correlation level. By enabling interactions among the relational representations of all query samples with respect to the support classes, QAM captures shared correlation patterns across queries and suppresses inconsistent or noisy responses. In this way, the generalized correlation matrix becomes more discriminative at the sample level.

3) Symmetrical Gated Integration: As the final step within the RCRG block, the SGI module is tasked with synergistically regulating the refined correlation matrices from the QAM branch ($\mathbf{M}_{\mathrm{QAM}}$) and the CGA branch ($\mathbf{M}_{\mathrm{CGA}}$). This integration is crucial for harmonizing the class-level priors from CGA with the sample-level enhancements from QAM. As depicted in Fig. 3(d), SGI employs a symmetrical gating strategy to dynamically and adaptively balance the contributions of these two complementary information streams. To implement this, we compute separate gate values to modulate their respective feature streams. Consequently, two gating units operate in parallel: a query-aware gate and a class-aware gate.

a) Query-Aware Gate: It is designed to leverage the querywise relational information distilled by the QAM branch to dynamically modulate the classwise correlation provided by the CGA branch. We start with the QAM-refined correlation matrix $\mathbf{M}_{\mathrm{QAM}} \in \mathbb{R}^{N_q \times N_c \times D}$, which deeply encodes the intrinsic dependencies among query samples. Based on this, our goal is to generate a dedicated gate value for each query sample, which is then used to adaptively filter and enhance the most valuable correlation patterns within $\mathbf{M}_{\mathrm{CGA}}$ for that specific query. Specifically, we first apply a max pooling operation to $\mathbf{M}_{\mathrm{QAM}}$ along the class dimension. This step distills a single, most salient D-dimensional correlation vector for each query sample. Then, we compute the mean over the feature dimension to aggregate information across all feature channels. Next, it is normalized to the range (0, 1) through a sigmoid function $\sigma(\cdot)$ to form the query-aware gate value. Finally, using this gate value, we perform elementwise multiplication with $\mathbf{M}_{\mathrm{CGA}}$, thereby achieving adaptive filtering at the query level. The entire process can be formulated as

$$\tilde{\mathbf{M}}_{\mathrm{CGA}} = \mathbf{M}_{\mathrm{CGA}} \odot \sigma\left(\frac{1}{D}\sum_{d=1}^{D}\mathrm{Max}_c\left(\mathbf{M}_{\mathrm{QAM}}(q,c,d)\right)\right) \tag{9}$$

where $\odot$ is the elementwise multiplication with broadcasting.

b) Class-Aware Gate: We aim to utilize the classwise information refined by the CGA branch to adaptively enhance the querywise correlation provided by the QAM branch. We start with the CGA-refined correlation matrix $\mathbf{M}_{\mathrm{CGA}} \in \mathbb{R}^{N_q \times N_c \times D}$, where the class-level interactions have been enhanced through the injection of global class priors. Based on this, our objective is to generate a dedicated gate value for each class, which, in turn, is used to amplify or suppress the querywise correlations within $\mathbf{M}_{\mathrm{QAM}}$ associated with that class. To produce more discriminative class-specific gate values, we first employ the SE attention [60] to further enhance the importance of each class channel within $\mathbf{M}_{\mathrm{CGA}}$. Specifically, we feed $\mathbf{M}_{\mathrm{CGA}}$ into an MLP to compress its

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feature dimension from D to 1, yielding an attention score map with a shape of $N_q \times N_c \times 1$. Subsequently, we apply a softmax function to this score map along the class dimension for normalization, followed by an elementwise multiplication with $\mathbf{M}_{\mathrm{CGA}}$. This "excitation" operation enables the model to adaptively recalibrate the feature responses of each class channel. Formally,

$$\hat{\mathbf{M}}_{\mathrm{CGA}} = \mathbf{M}_{\mathrm{CGA}} \odot \mathrm{Softmax}\left(\mathrm{MLP}\left(\mathbf{M}_{\mathrm{CGA}}\right)\right). \tag{10}$$

Building upon $\hat{\mathbf{M}}_{\mathrm{CGA}}$, we generate the class-aware gate value through the following steps. First, the matrix is summed along the query dimension to aggregate the overall responses of all query samples for each class. Next, global averaging is applied to compress the resulting D-dimensional feature vectors into scalars. Finally, these scalars are passed through a sigmoid function to obtain the class-aware gate value. Using these gate values, we perform an elementwise multiplication on $\mathbf{M}_{\mathrm{QAM}}$ with broadcasting, thereby achieving a class-level modulation. This process can be expressed as

$$\tilde{\mathbf{M}}_{\mathrm{QAM}} = \mathbf{M}_{\mathrm{QAM}} \odot \sigma\left(\frac{1}{D}\sum_{d=1}^{D}\sum_{q=1}^{N_q}\hat{\mathbf{M}}_{\mathrm{CGA}}\left(q,c,d\right)\right). \tag{11}$$

Finally, $\tilde{\mathbf{M}}_{\mathrm{CGA}}$ and $\tilde{\mathbf{M}}_{\mathrm{QAM}}$ are summed and then passed through an MLP to synergistically fuse the complementary classwise and querywise information, given by

$$\mathbf{M}_{\mathrm{SGI}} = \tilde{\mathbf{M}}_{\mathrm{CGA}} + \tilde{\mathbf{M}}_{\mathrm{QAM}}. \tag{12}$$

Note that the output $\mathbf{M}_{\mathrm{SGI}} \in \mathbb{R}^{N_q \times N_c \times D}$ can be viewed as $\tilde{\mathbf{M}}^{l}_{\mathrm{GCM}}$ in (3), which results from refining $\mathbf{M}^{l-1}_{\mathrm{GCM}}$ through an RCRG block.

E. Correlation Projection

After L stages of progressive refinement within the RCRG blocks, we obtain the final refined generalized correlation matrix as $\mathbf{M}^{L}_{\mathrm{GCM}} \in \mathbb{R}^{N_q \times N_c \times D}$. This matrix encapsulates the highly discriminative relational information between each query sample and all support class prototypes. To derive class predictions from this structured representation, a simple yet effective correlation projection module, implemented as an MLP, is used to produce the final classification logits. It comprises two linear layers with an intermediate ReLU activation function and a dropout layer for regularization. This MLP maps each entry in the D-dimensional generalized correlation matrix to a single scalar value. This process yields a logit tensor $\mathbf{M}_{\mathrm{LGT}} \in \mathbb{R}^{N_q \times N_c}$, where each entry $\mathbf{M}_{\mathrm{LGT}}(q,c)$ represents the predicted score for the qth query sample belonging to the cth class.

The PCRNet is trained end-to-end by minimizing the standard cross-entropy loss between the predicted logits and the ground-truth labels of the query samples

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{N_q}\sum_{q=1}^{N_q}\log\left(\frac{\exp\left(\mathbf{M}_{\mathrm{LGT}}\left(q,\hat{c}\right)\right)}{\sum_{c=1}^{N_c}\exp\left(\mathbf{M}_{\mathrm{LGT}}\left(q,c\right)\right)}\right) \tag{13}$$

where $\hat{c}$ is the true label for the qth query sample.

IV. Experiments

A. Datasets

The datasets encompass a source domain (SD) and a TD. The SD comprises classes with abundant labeled samples, allowing us to learn rich prior knowledge. This knowledge is then leveraged to enhance the classification of novel classes within the TD, for which only a few labeled samples are available. Following [10], [11], and [12], we employ the Chikusei[1] (CKS) dataset as the SD. Furthermore, the TD contains four widely used benchmark datasets: the University of Pavia[2] (UP), Salinas[2] (SA), Indian Pines[2] (IP), and WHU-Hi-HanChuan[3] (HC). Detailed information for each land-cover category in the SD and TD datasets is summarized in Table I. All experimental analyses are performed on the TD datasets.

1) SD Dataset: CKS Dataset: Originating from agricultural and urban terrains in Chikusei, Japan, the CKS dataset was captured by the Headwall Hyperspec-VNIR-C sensor, offering a spatial resolution of 2.5 m. This dataset features 128 spectral bands that span the 363~1018 nm wavelength range, presented as an image of 2517 × 2335 pixels. It further provides 77 592 labeled samples distributed across 19 distinct land-cover categories.

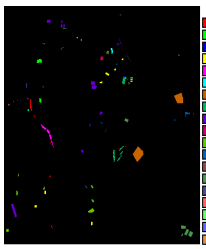
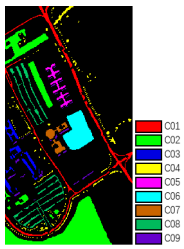
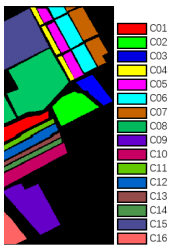
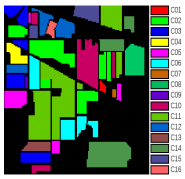
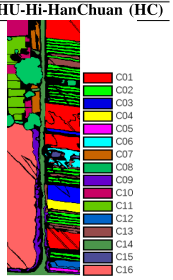
2) TD Datasets:

1) *UP Dataset:* Acquired over the UP in northern Italy, this dataset was collected using the ROSIS sensor, yielding a spatial resolution of 1.3 m. It consists of 103 spectral bands and furnishes 42 776 labeled samples classified into nine categories, all within an image area of 610 × 610 pixels. The scene is predominantly characterized by urban elements, including buildings, roadways, and other man-made structures.
2) *SA Dataset:* Gathered from Salinas, California, via the AVIRIS sensor, this dataset provides a spatial resolution of 3.7 m. While the raw data initially contained 224 spectral bands covering 400~2500 nm, 204 bands were retained for analysis after the removal of bands compromised by water absorption and calibration issues. This dataset offers 54 129 labeled samples across 16 land-cover categories. The image resolution is 512 × 217. The landscape is largely composed of various vegetable crops, exposed soil, and botanical gardens.
3) *IP Dataset:* Captured over Indiana using the AVIRIS sensor, the IP dataset presents a spatial resolution of 20 m. From an initial set of 224 spectral bands covering 400~2500 nm, 200 bands were selected post-exclusion of uncalibrated and noisy bands affected by water absorption. The dataset includes 10 249 labeled instances across 16 categories, with an image resolution of 145 × 145 pixels. The landscape is primarily agricultural approximately two-thirds, with the remainder being forests or natural perennial vegetation.
4) *HC Dataset:* This dataset was acquired in Hanchuan, Hubei province, China, with a Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot

[1]Online: https://naotoyokoya.com/Download.html
[2]Online: https://rslab.ut.ac.ir/data
[3]Online: https://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE I
GT AND THE CORRESPONDING CLASS DETAILS OF SOURCE AND TD DATASETS

Chikusei (CKS)			Pavia University (UP)		Salinas (SA)		Indian Pines (IP)		WHU-Hi-HanChuan (HC)	
										
ID	**land-covers**	**samples**	**land-covers**	**samples**	**land-covers**	**samples**	**land-covers**	**samples**	**land-covers**	**samples**
C01	Water	2845	Asphalt	6631	Brocoli-green-weeds-1	2009	Asphalt	46	Strawberry	44735
C02	Bare soil(school)	2859	Meadows	18649	Brocoli-green-weeds-2	3726	Corn-notill	1428	Cowpea	22753
C03	Bare soil(park)	286	Gravel	2099	Fallow	1976	Corn-mintill	830	Soybean	10287
C04	Bare soil(farmland)	48525	Trees	3064	Fallow-rough-plowees	1394	Corn	237	Sorghum	5353
C05	Natural plants	4297	Painted metal sheets	1345	Fallow-smooth	2678	Grass-pasture	483	Water spinach	1200
C06	Weeds in farmland	1108	Bare Soil	5029	Stubble	3959	Grass-trees	730	Stubble	4533
C07	Forest	20516	Bitumen	1330	Celery	3579	Grass-pasture-mowed	28	Greens	5903
C08	Grass	6515	Self-Blocking Bricks	3682	Grapes-untrained	11271	Hay-windrowed	478	Trees	17978
C09	Rice field(grown)	13369	Shadows	947	Soil-vinyard-develop	6203	Oats	20	Grass	9469
C10	Rice field(first stage)	1268	**Total**	42776	Corn-senesced-green-weeds	3278	Soybean-notill	971	Red roof	10516
C11	Row crops	5961			Lettuce-romaine-4wk	1068	Soybean-mintill	2455	Gray roof	16911
C12	Plastic house	2193			Lettuce-romaine-5wk	1927	Soybean-clean	593	Plastic	3679
C13	Manmade(non-dark)	1220			Lettuce-romaine-6wk	916	Wheat	205	Bare soil	9116
C14	Manmade(dark)	7664			Lettuce-romaine-7wk	1070	Woods	1265	Road	18560
C15	Manmade(blue)	431			Vinyard-untrained	7268	Buildings-Grass-Trees-Drives	386	Bright object	1136
C16	Manmade(red)	222			Vinyard-vertical-trellis	1807	Stone-Steel-Towers	93	Water	75401
C17	ManMade grass	1040			**Total**	54129	**Total**	10249	**Total**	257450
C18	Asphalt	801								
C19	Paved ground	145								
	Total	77592								

X6 UAV V1 platform. The UAV operated at an altitude of 250 m. The dataset covers a rural–urban fringe area and comprises 38 622 labeled samples distributed across 16 land-cover categories. These categories include general classes such as buildings and water bodies, as well as seven specific types of cultivated land: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and various greens. It comprises 274 spectral bands covering 400~1000 nm and boasts a high spatial resolution of approximately 0.109 m. The image resolution is 1217 × 303. A key feature of this dataset is the extensive presence of shadows resulting from afternoon data acquisition.

B. Experimental Details

We leverage the rich prior knowledge from the SD dataset to learn a generalizable model that can be adapted to the TD dataset. Our approach is built upon an episode-based metalearning framework, simulating C-way K-shot classification tasks. To ensure consistency in the number of classes between the SD and TD datasets for learning the global class prior, C classes are randomly selected from the SD dataset, where C corresponds to the number of classes present in the TD dataset. Both the SD and TD datasets maintain a CGMU, respectively. Specifically, C is configured to 9 for the UP dataset and 16 for the SA, IP, and HC datasets. Unless otherwise stated, K is set to 5 to maintain a five-shot setting during training and evaluation. In line with prior FHSIC works [12], [59], each episode consists of $N = 19$ query samples per class, and the input image patch is set to 9×9 pixels (i.e., $P = 9$). The dimension of the feature embedding is set to 128 (i.e., $D = 128$). MICD [12] is utilized as our feature embedding module for all experiments. We train the network model using the Adam optimizer with a learning rate of 0.001. The total number of training episodes is empirically set to 10 000, which helps the model generalize effectively to the novel class with limited labels. To ensure the reliability and statistical significance of our findings, all results are averaged over ten independent experimental runs. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU.

To evaluate classification performance, we employ four standard metrics established in the FHSIC task: class-specific accuracy (CA), average accuracy (AA), overall accuracy (OA), and the kappa coefficient (Kappa). These metrics provide a comprehensive assessment of the model's classification efficacy, where higher values indicate better performance.

C. Comparison With Existing Methods

To validate the effectiveness of our proposed method, we compare it with other state-of-the-art FHSIC approaches on TD datasets as follows.

1) *Fast3DCNN [20]* integrated a 3-D CNN to jointly extract spatial and spectral features.
2) *DCFSL [48]* presented a unified framework with a conditional adversarial domain adaptation strategy to reduce domain shift.
3) *Gia-CFSL [61]* promoted domain transfer by aligning cross-domain graph information at both feature and distribution levels.
4) *DM-MRN [44]* extended the conventional RN with a dual-metric, multilevel architecture, aiming to learn more discriminative features through enriched task diversity.
5) *DA-CAA [9]* proposed distribution-aware and class-adaptive aggregation modules to transfer class-level information from base classes to novel classes;
6) *CTF-SSCL [10]* designed a CNN-Transformer network with semisupervised contrastive learning to improve the

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TABLE II
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE UP DATASET

No.	Fast3DCNN [20]	DCFSL [48]	Gia-CFSL [61]	DM-MRN [44]	DA-CAA [9]	CTF-SSCL [10]	FDFSL [12]	TEFSL [59]	**Ours**
1	68.17±9.85	76.07±10.09	79.99±13.67	79.36±8.51	81.98±7.19	**88.32±5.62**	76.16±10.22	85.01±3.27	88.29±4.67
2	81.14±9.45	83.71±7.59	82.73±8.62	75.91±9.81	86.22±6.93	77.73±10.82	**87.95±7.97**	80.25±11.69	85.42±7.95
3	56.72±11.23	57.44±12.94	64.27±10.43	**79.37±8.41**	72.73±10.66	68.51±10.19	69.80±8.56	66.58±13.14	78.36±11.76
4	91.63±3.83	90.13±5.52	92.51±2.94	88.84±10.06	93.31±3.29	89.77±3.19	**93.84±5.05**	90.49±2.35	89.15±5.14
5	98.25±3.17	98.01±3.07	99.45±0.60	99.78±0.35	99.77±0.31	99.81±0.25	**99.84±0.29**	99.38±0.65	99.70±0.36
6	59.38±11.32	68.40±12.88	76.52±12.95	**85.62±16.01**	70.04±11.72	75.68±12.48	64.03±12.40	69.46±12.70	75.96±15.31
7	82.94±6.94	75.90±9.95	83.06±6.88	**98.92±0.61**	88.32±5.94	92.39±7.57	87.98±6.28	90.20±4.33	95.18±3.04
8	64.98±18.54	67.61±13.29	77.13±13.30	75.75±10.39	84.00±11.56	82.73±10.83	72.10±12.92	**85.85±8.68**	81.80±11.98
9	96.18±3.37	98.18±1.86	98.76±1.85	**99.38±0.51**	96.49±4.25	93.87±7.66	99.24±1.31	95.35±2.91	94.65±5.62
AA	77.71±1.51	79.49±3.06	83.82±2.09	87.17±2.45	85.87±2.50	85.42±2.02	83.44±2.23	84.73±1.77	**87.61±1.91**
OA	75.66±3.78	79.03±2.99	81.77±4.31	77.30±5.91	84.03±2.67	81.48±4.19	82.10±2.94	81.50±4.31	**85.32±2.65**
Kappa	68.59±4.19	72.83±3.54	76.57±4.98	71.82±6.77	79.24±3.20	76.37±4.77	76.61±3.49	76.23±4.91	**80.93±3.11**

TABLE III
CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE SA DATASET

No.	Fast3DCNN [20]	DCFSL [48]	Gia-CFSL [61]	DM-MRN [44]	DA-CAA [9]	CTF-SSCL [10]	FDFSL [12]	TEFSL [59]	**Ours**
1	99.48±0.69	99.01±1.02	99.62±0.33	**99.88±0.24**	98.86±1.31	98.95±2.11	99.02±1.39	98.63±3.12	99.11±1.07
2	99.76±0.63	99.16±1.08	99.18±1.23	**99.86±0.17**	99.59±0.44	99.77±0.47	96.84±3.57	99.55±1.14	98.99±1.51
3	94.02±6.40	91.81±8.20	93.60±9.05	93.38±14.10	89.67± 9.80	**95.01±7.04**	89.25±12.69	80.26±17.00	93.56±9.59
4	99.60±0.27	**99.67±0.38**	99.11±1.40	99.12±1.71	99.53±0.55	99.10±1.22	99.55±0.54	99.27±0.55	98.56±1.52
5	92.34±3.26	91.10±3.86	92.71±3.67	95.20±4.95	94.38±4.34	92.10±4.66	95.46±1.12	94.12±5.47	**96.18±1.36**
6	98.22±0.92	99.10±0.79	98.09±2.58	99.40±0.91	**99.55±0.78**	99.25±1.04	99.43±0.59	99.27±0.94	99.39±0.80
7	**99.72±0.43**	98.59±2.41	97.93±4.09	99.02±2.47	99.12±0.96	99.70±0.22	98.74±1.87	98.52±2.28	98.31±2.98
8	69.00±14.31	**77.98±9.03**	73.15±11.71	65.52±12.55	77.56±9.58	73.50±10.14	69.56±13.45	76.89±9.57	76.92±9.76
9	94.62±2.87	97.48±2.72	98.43±2.56	**99.88±0.24**	98.61±1.48	98.78±1.58	99.17±0.89	98.57±1.88	99.75±0.45
10	83.52±5.33	76.62±13.06	82.57±6.77	**89.53±8.54**	85.62±12.38	81.39±12.92	77.55±16.32	88.65±3.87	86.72±7.93
11	95.73±4.96	98.25±0.75	97.34±2.05	98.27±2.38	**98.52±1.09**	97.44±1.50	96.15±4.75	98.13±1.72	98.34±1.15
12	99.73±0.44	99.24±0.71	**99.77±0.27**	97.60±1.89	99.44±0.78	98.23±3.26	99.46±1.06	99.29±0.86	99.32±1.81
13	97.84±2.21	98.70±1.56	99.05±1.02	98.91±1.11	99.32±0.71	98.32±2.28	**99.70±0.27**	99.17±1.50	99.26±0.83
14	98.67±0.58	97.57±2.32	98.83±0.77	**99.28±0.52**	98.34±0.74	96.50±3.95	98.55±1.24	97.05±4.31	99.12±1.24
15	63.21±14.24	74.74±5.04	77.82±8.59	74.35±10.60	78.32±8.58	79.78±8.26	73.44±9.36	79.09±5.54	**79.96±6.21**
16	88.02±7.39	88.18±8.42	88.71±6.44	**97.65±1.19**	90.51±7.11	90.82±7.24	89.70±4.22	92.89±6.75	93.84±5.10
AA	92.09±1.31	92.95±0.87	93.50±1.18	94.18±0.99	94.18±0.87	93.66±1.47	92.60±1.45	93.71±0.93	**94.83±1.10**
OA	85.63±2.55	88.78±1.80	88.75±2.29	87.86±1.48	90.14±1.17	89.25±2.32	87.15±2.17	89.89±2.13	**90.65±2.31**
Kappa	84.05±2.80	87.54±1.98	87.52±2.52	86.54±1.61	89.05±1.28	88.08±2.55	85.74±2.38	88.77±2.34	**89.62±2.55**

classification performance under the scarcity of labeled samples.

7) *FDFSL [12]* addressed the discrepancy between the SD and TD through feature disentanglement techniques and a central position encoding module while employing a self-distillation strategy to enrich feature representation.
8) *TEFSL [59]* employed a transductive FSL framework with enhanced spectral–spatial embedding to make full use of the limited available prior information. While the above methods mainly focus on improving classification performance through feature refinement, alignment, or enhancement under few-shot settings, our approach differs by explicitly modeling and progressively refining query–class correlations. Tables II–V report the mean accuracy and standard deviation of CA, OA, AA, and Kappa across the four TD datasets, all evaluated under the five-shot setting.

Table II presents the classification performance of various methods on the UP dataset. It is evident that our proposed approach demonstrates superior efficacy, achieving AA, OA, and Kappa values of 87.61%, 85.32%, and 80.93%, respectively. Notably, our approach significantly outperforms several state-of-the-art cross-domain FHSIC methods. Specifically, compared to DA-CAA [9], FDFSL [12], and TEFSL [59], our approach yields OA improvements of 1.53%, 3.92%, and 4.68%, respectively. Similarly, Table III summarizes the classification results on the SA dataset under the five-shot condition. Our approach achieves the best result for FHSIC, achieving AA, OA, and Kappa values of 94.83%, 90.65%, and 89.62%. An examination of the CA results in both Tables II and III reveals that our approach consistently achieves more robust and balanced predictions across most categories. This consistent superiority suggests that our CR paradigm effectively captures and utilizes structured interaction relationships among samples from a holistic perspective.

We further validate our method on the more challenging IP and HC datasets, which are characterized by significant class imbalance and high spectral similarity. On the IP dataset, as shown in Table IV, our PCRNet achieves the AA, OA, and Kappa values of 83.99%, 74.11%, and 70.81%, respectively. These results represent notable improvements of 2.25% in AA, 4.26% in OA, and 4.99% in Kappa over the state-of-the-art method CTF-SSCL [10]. Notably, when analyzing the CA performance, our approach secures the best and second-best results for ten out of 16 categories, surpassing all compared methods. For instance, for Class #2 (“Corn-notill”), Class #12 (“Soybean-clean”), and Class #15 (“Buildings-Grass-Trees-Drives”), our approach achieved CA that was 4.75%, 4.24%, and 5.44% higher than the second-best results, respectively. This improved performance is also verified on the HC dataset, as shown in Table V. Our approach achieves leading performance with AA, OA, and Kappa values of 72.52%, 75.36%, and 71.66%, respectively, surpassing the second-best method, CTF-SSCL [10], by 3.74% in AA, 3.85% in OA, and 4.62% in Kappa. Similarly, in terms of CA, our method ranks either first or second in a remarkable 11 out of the 16 categories. For

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TABLE IV

CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE IP DATASET

No.	Fast3DCNN [20]	DCFSL [48]	Gia-CFSL [61]	DM-MRN [44]	DA-CAA [9]	CTF-SSCL [10]	FDFSL [12]	TEFSL [59]	**Ours**
1	97.32±7.27	92.44±12.83	92.44±13.38	87.07±12.62	94.88±9.78	**98.54±1.95**	94.63±12.90	98.05±3.05	97.80±2.43
2	45.37±11.25	38.92±7.94	49.73±9.32	45.57±10.25	53.71±13.20	49.47±10.96	46.49±9.93	61.79±9.03	**66.54±10.79**
3	51.70±9.52	45.57±5.84	53.38±9.81	46.73±10.87	58.13±5.72	**62.53±8.32**	58.13±9.59	60.42±8.21	61.93±9.32
4	67.28±14.11	67.50±14.90	81.51±13.13	84.13±13.35	81.47±11.36	87.20±14.32	76.98±15.87	87.67±8.77	**89.01±8.93**
5	66.21±10.43	67.82±6.48	75.65±7.88	72.86±6.40	76.84±8.99	78.99±8.94	70.36±9.37	75.65±8.27	**80.90±7.96**
6	87.77±8.85	76.68±8.12	82.87±10.12	**94.27±3.50**	87.03±5.64	89.72±4.30	86.56±9.57	85.88±7.08	91.34±5.51
7	**100.00±0.00**	98.26±3.98	99.13±2.61	98.26±3.98	99.56±1.30	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	90.74±10.78	88.14±8.79	92.60±5.31	83.91±11.00	**94.31±7.37**	93.49±7.11	89.20±13.56	88.82±11.21	91.31±13.20
9	**100.00±0.00**	**100.00±0.00**	99.33±2.00	**100.00±0.00**	98.67±4.00	99.33±2.00	**100.00±0.00**	**100.00±0.00**	98.67±2.81
10	55.90±12.45	61.59±10.80	63.06±8.75	66.70±6.23	66.02±8.45	**68.11±10.88**	60.51±10.76	64.99±10.53	66.67±11.63
11	54.16±16.59	59.19±12.79	60.90±13.45	57.26±14.02	59.70± 9.82	**64.47±12.07**	60.71±17.39	59.50±10.36	62.98±12.35
12	48.89±11.83	35.58±8.37	49.05±11.66	41.24±7.83	56.92±14.13	53.88±14.21	46.94±12.18	56.51±12.52	**61.16±16.68**
13	98.80±1.76	98.20±2.99	99.25±1.35	**99.40±1.04**	98.20±2.30	99.35±0.90	98.05±3.12	98.45±1.59	97.00±3.19
14	81.33±10.72	83.06±10.51	81.49±8.52	**93.55±5.59**	84.66±10.13	87.30±6.21	84.51±11.09	83.63±11.50	91.10±7.64
15	74.33±7.13	56.88±10.73	68.58±12.04	61.36±8.47	83.17±10.73	83.25±9.02	76.09±11.92	82.73±14.03	**88.69±6.89**
16	99.09±0.99	98.98±2.35	98.98±1.72	**99.65±0.52**	99.54±0.75	98.52±2.88	99.32±1.70	98.07±4.04	98.75±3.19
AA	76.18±1.37	73.05±1.73	78.00±2.29	77.00±2.25	80.80±0.95	82.14±2.03	78.03±1.00	81.39±1.18	**83.99±2.30**
OA	63.29±2.86	61.57±2.77	66.64±3.97	66.09±3.22	69.40±1.96	71.08±4.11	66.56±3.09	70.16±2.73	**74.11±3.98**
Kappa	58.65±2.94	56.59±2.96	62.44±4.23	61.98±3.34	65.51±2.19	67.44±4.49	62.27±3.17	66.49±2.87	**70.81±4.33**

TABLE V

CLASSIFICATION RESULTS (%) OF VARIOUS METHODS WITH FIVE LABELED SAMPLES PER CLASS ON THE HC DATASET

No.	Fast3DCNN [20]	DCFSL [48]	Gia-CFSL [61]	DM-MRN [44]	DA-CAA [9]	CTF-SSCL [10]	FDFSL [12]	TEFSL [59]	**Ours**
1	74.68±8.54	77.95±9.76	80.31±9.85	59.85±5.54	78.03±7.55	76.41±6.73	79.49±6.79	**81.44±7.65**	79.77±9.81
2	32.44±12.76	40.66±13.66	39.79±11.66	31.13±8.53	47.67±11.77	42.69±9.93	43.13±11.79	47.83±8.87	**57.64±11.63**
3	59.69±12.64	61.50±10.12	73.96±8.53	35.46±10.05	73.56±8.24	**82.39±7.11**	77.93±7.74	79.83±8.96	81.21±7.27
4	92.57±3.87	91.74±3.80	91.11±8.74	79.21±11.99	93.86±2.98	**94.86±4.49**	94.67±2.21	93.75±2.51	94.44±2.95
5	79.78±9.64	83.43±7.46	84.05±8.62	85.16±12.46	85.51±10.21	86.88±13.60	91.23±8.64	88.61±7.31	**93.33±5.65**
6	24.08±5.92	30.25±7.90	36.80±10.87	24.81±8.87	38.55±8.97	**49.72±13.80**	37.38±8.02	36.89±7.63	42.11±11.11
7	82.67±9.12	86.05±6.88	75.93±13.69	86.52±8.00	76.87±13.23	82.14±10.73	**88.56±2.68**	78.08±11.91	76.23±16.86
8	39.10±11.70	38.63±9.13	42.67±8.26	41.79±14.32	46.84±11.02	51.05±11.03	44.64±12.15	44.43±8.11	**56.58±8.33**
9	26.70±11.43	27.24±3.98	39.04±13.13	25.85±7.58	45.42±15.33	47.89±16.68	33.11±8.11	37.62±10.63	**65.90±17.59**
10	74.86±8.84	76.77±7.34	78.99±10.03	75.04±10.92	87.38±6.69	89.99±8.86	79.62±7.30	78.91±15.14	**94.35±4.94**
11	61.24±17.55	63.68±17.86	57.15±14.41	69.45±14.60	63.40±14.89	**75.65±15.26**	58.49±11.60	54.27±13.82	68.56±13.40
12	41.11±18.84	51.87±12.81	55.20±13.77	52.25±10.06	59.12±18.02	57.29±12.51	57.41±10.76	60.11±15.28	**66.30±17.98**
13	27.14±15.44	35.33±12.96	43.06±12.40	18.34±8.33	37.61±6.74	**45.90±11.57**	31.74±12.09	35.81±7.61	44.27±10.51
14	61.39±17.03	62.56±12.40	61.02±12.03	45.78±9.63	55.35±15.14	62.23±12.77	**65.31±13.48**	51.30±16.27	65.30±16.40
15	75.05±16.49	71.48±11.46	69.00±9.88	76.15±12.26	72.94±13.98	85.05±7.95	65.68±13.46	71.64±15.57	**85.76±12.22**
16	85.77±10.91	88.42±11.47	89.67±10.68	89.12±10.35	**90.49±9.63**	88.20±10.88	87.79±11.05	88.79±11.43	88.63±10.97
AA	58.64±3.29	61.72±2.49	63.61±3.37	55.99±1.71	65.79±1.65	69.90±2.11	64.76±2.39	64.33±2.22	**72.52±2.15**
OA	65.05±3.49	68.10±3.33	69.76±3.38	61.64±3.45	71.15±2.58	72.56±3.11	69.88±2.29	69.77±3.63	**75.36±4.41**
Kappa	59.88±3.80	63.27±3.65	65.13±3.68	56.08±3.56	66.77±2.69	68.49±3.28	65.31±2.40	65.14± 3.86	**71.66±4.76**

Class #2 ("Cowpea"), Class #8 ("Trees"), Class #9 ("Grass"), and Class #12 ("Plastic"), our approach outperformed the second-best results by substantial margins of 9.81%, 5.53%, 18.01%, and 6.19%, respectively. The superior performance on the IP and HC datasets, which are characterized by inherent class imbalance and challenging scene features, fully validates the superiority and robustness of our proposed PCRNet.

To provide an intuitive comparison of classification performance, we visualize the classification maps generated by our PCRNet and the competing FHSIC methods on the TD datasets. These visual results, presented in Figs. 4–7, are displayed alongside their corresponding false-color images and ground-truth (GT) maps for reference. A consistent observation across all datasets is that our approach produces visually clear classification maps, exhibits greater homogeneity within classified regions, and demonstrates higher fidelity to the GT. Specifically, as displayed in Fig. 4 for the UP dataset, our PCRNet achieves a notably more precise and coherent classification. For instance, areas corresponding to "Meadows" (Class #2), "Trees" (Class #4), and "Self-Blocking Bricks" (Class #8) are delineated with higher accuracy and fewer misclassified pixels compared to other methods. This indicates our approach's effectiveness in distinguishing diverse urban land cover types with complex spatial arrangements. Turning to Fig. 5 for the SA dataset, many competing methods exhibit a tendency to confuse spectrally similar agricultural classes, particularly "Grapes-untrained" (Class #8) and "Vineyard-untrained" (Class #15), resulting in noisy and fragmented classification maps. In contrast, our method demonstrates a superior capability to classify these classes, yielding spatially consistent and contiguous regions that more closely match the GT. This advantage is particularly evident in the more complex scenes. As shown in Figs. 6 and 7, our PCRNet delivers classification maps characterized by higher overall clarity, smoother classified regions, and more well-defined boundaries. Specifically, on the HC dataset, classes with highly similar spectral profiles, such as "Cowpea" (Class #2) and "Grass" (Class #9), are rendered with noticeably clearer boundaries. This superior performance, particularly in differentiating spectrally similar classes within complex scenes, underscores the robust modeling capacity of our proposed CR paradigm to capture relational information between support and query samples.

In addition to the quantitative and qualitative comparisons presented under the five-shot condition, we further conduct a comprehensive evaluation of FHSIC performance across various methods under a spectrum of shot settings (i.e., $K \in \{1, 2, 3, 4, 5\}$). As illustrated in Fig. 8, a general trend

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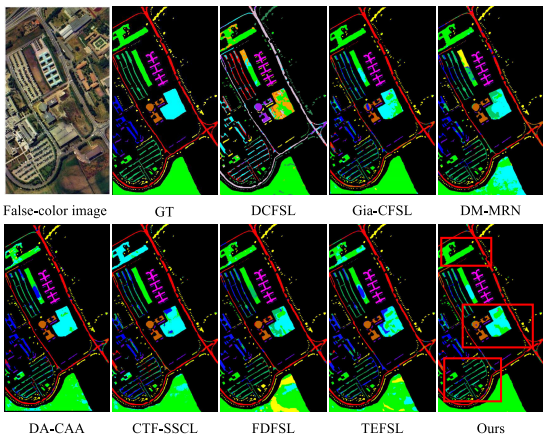


Fig. 4. Visual comparisons against existing methods on the UP dataset.

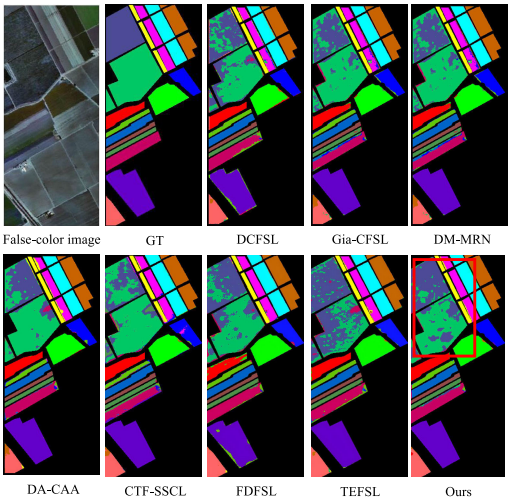


Fig. 5. Visual comparisons against existing methods on the SA dataset.

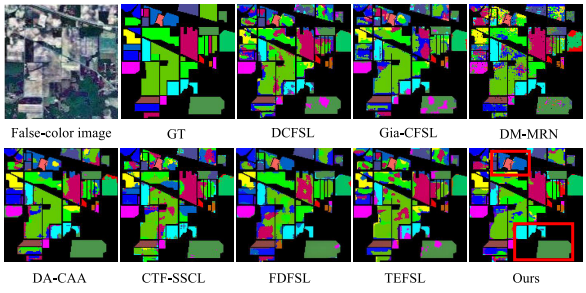


Fig. 6. Visual comparisons against existing methods on the IP dataset.

is observed where the classification accuracy of all evaluated algorithms improves with an increasing number of labeled samples per class across the four TD datasets. Note that our approach consistently outperforms competing methods in almost all cases. This sustained superiority can essentially be attributed to PCRNet's core CR paradigm. By explicitly modeling and optimizing the structured correlation matrix rather than relying solely on individual feature learning, our PCRNet is better equipped to generalize from limited data.

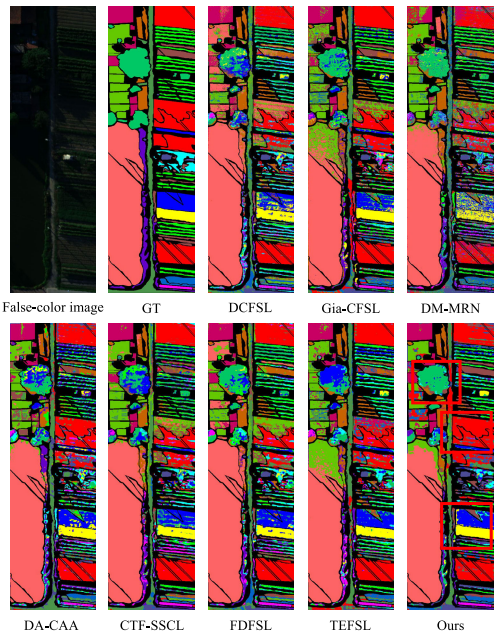


Fig. 7. Visual comparisons against existing methods on the HC dataset.

D. Ablation Studies

We conduct comprehensive ablation studies to evaluate the influence of individual components in PCRNet. Additional ablation studies are provided in Section I of the supplementary material, including detailed analyses of the MGA mechanism and the feature channel dimension.

1) Ablation on the Number of RCRG Block: To investigate the tradeoff between performance and model complexity, we analyzed the impact of the RCRG block count (L) ranging from 1 to 5. As illustrated in Fig. 9, accuracy improves consistently as L increases from 1 to 3, validating the efficacy of progressive refinement in structuring relational information. However, increasing L beyond 3 yields diminishing returns or even performance drops on datasets such as UP and HC, attributed to the increased overfitting risk inherent in few-shot scenarios. In terms of the computational complexity, the parameters grow linearly with network depth (specifically 0.53M, 0.76M, 0.98M, 1.21M, and 1.44M for $L = 1$–5, respectively). As such, we select $L = 3$ as the optimal configuration to balance robust classification accuracy with computational efficiency.

2) Ablation on Key Components in RCRG Blocks: Each RCRG block has three key modules: the CGA module with the MGA mechanism for introducing global class commonalities, the QAM for refining querywise correlations, and the SGI module for fusing correlation clues. Therefore, we conduct an ablation study on the four TD datasets to validate the contribution of the three key components, as presented in Table VI. The baseline model is established by removing all RCRG blocks from our PCRNet, i.e., without any CR process. As can be seen, simply introducing some refinement to the generalized correlation matrix is sufficient to substantially increase the HSIC performance beyond the baseline, even if the refinement mechanism itself is incomplete (e.g., by using an RCRG with one of its components ablated). Moreover,

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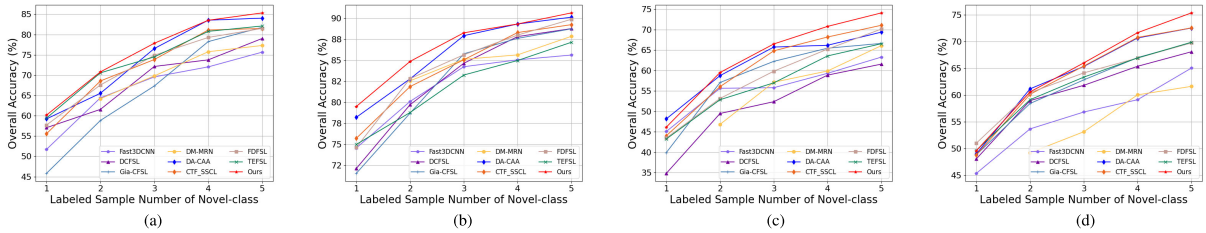


Fig. 8. Comparison of classification performance (%) of various methods from one-shot to five-shot on the TD datasets. Our approach demonstrates outstanding classification performance across various few-shot conditions (a) The UP dataset. (b) The SA dataset. (c) The IP dataset. (d) The HC dataset.

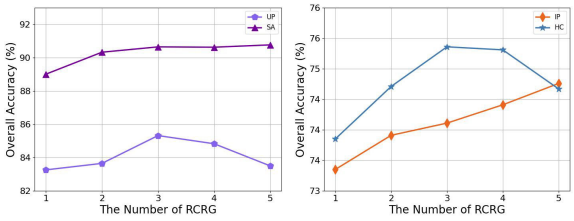


Fig. 9. Ablation comparisons (%) with different numbers of RCRG blocks (L) on the TD datasets. Left: performance comparison on the UP and SA datasets. Right: performance comparison on the IP and HC datasets.

TABLE VI
ABLATION COMPARISONS (%) OF KEY COMPONENTS IN RCRG BLOCKS ON THE TD DATASETS

	Baseline	CGA	QAM	SGI	AA	OA	Kappa
UP	✓				83.11±2.43	81.01±4.11	75.39±4.86
	✓		✓	✓	86.90±2.85	84.15±3.58	79.54±4.18
	✓	✓		✓	86.36±1.81	83.82±3.12	79.19±3.68
	✓	✓	✓		86.90±2.18	84.40±4.03	79.94±4.73
	✓	✓	✓	✓	**87.61±1.91**	**85.32±2.65**	**80.93±3.11**
SA	✓				92.92±2.11	89.41±2.82	88.25±3.12
	✓		✓	✓	94.50±1.14	90.47±2.81	89.56±3.09
	✓	✓		✓	94.15±1.21	90.03±2.44	89.01±2.69
	✓	✓	✓		94.47±1.38	89.91±2.85	89.48±3.13
	✓	✓	✓	✓	**94.83±1.10**	**90.65±2.31**	**89.62±2.55**
IP	✓				77.60±2.02	66.88±4.40	62.87±4.75
	✓		✓	✓	83.43±1.66	73.55±3.76	70.30±4.04
	✓	✓		✓	83.05±1.89	73.70±3.03	70.25±3.24
	✓	✓	✓		83.60±2.37	73.79±4.46	70.35±4.85
	✓	✓	✓	✓	**83.99±2.30**	**74.11±3.98**	**70.81±4.33**
HC	✓				66.37±2.75	71.44±4.91	67.16±5.27
	✓		✓	✓	71.29±3.46	74.66±3.63	70.87±3.93
	✓	✓		✓	71.87±2.45	74.63±4.58	70.88±4.95
	✓	✓	✓		72.26±3.25	74.87±4.49	71.13±4.90
	✓	✓	✓	✓	**72.52±2.15**	**75.36±4.41**	**71.66±4.76**

CGA, QAM, and SGI each make a substantial contribution to the final performance improvement. This indicates their positive role in refining the generalized correlation matrix to be more discriminative.

E. Effectiveness Evaluation of the Paradigm Shift

A cornerstone of our contribution is the departure from the conventional "representation-metric" framework that underpins most existing FHSIC methods, including those based on RNs. Instead of refining individual sample features, our proposed PCRNet pioneers a "relation-reasoning" framework. This new approach explicitly treats the query–class relationships as a holistic, structured entity and iteratively optimizes this entire correlation structure.

To evaluate the effectiveness of our proposed paradigm, we conducted a comparative experiment. We establish a baseline that represents the conventional feature refinement paradigm by removing all RCRG blocks from our full PCRNet architecture while keeping the feature embedding backbone identical for a fair comparison. This ablation effectively reduces our model to a structure that mimics RN-based methods: it performs a one-step mapping from isolated feature pairs to a final scalar relation score, devoid of any iterative reasoning or contextual refinement. We refer to our full model as the CR paradigm and this baseline as the CR paradigm for clarity. Furthermore, to demonstrate the scalability and versatility of our PCRNet, we integrate our CR paradigm with three distinct feature embedding backbones: MICD [12], CMFSL [8], and Fast3DCNN [20].

As illustrated in Fig. 10, across all datasets and for every backbone, our proposed CR paradigm consistently and substantially outperforms the FR (i.e., RN-based) baseline. This improvement provides strong evidence for our central thesis: shifting the model's core operation from simply measuring isolated features to progressively reasoning over their holistic relationships unlocks a new level of performance in FHSIC. Note that this trend holds regardless of the chosen feature embedding backbone. As detailed in Fig. 10, when using MICD [12], CMFSL [8], and Fast3DCNN [20] as backbones, the OAs of our CR paradigm on the UP dataset are 85.32%, 83.23%, and 78.29%, respectively. This pattern of MICD achieving the highest performance continues on the SA dataset (OAs of 90.65%, 89.95%, and 87.43%) and the IP dataset (OAs of 74.11%, 72.63%, and 65.15%). This analysis not only validates the scalability of our PCRNet but also reinforces that the observed performance gains are intrinsic to the paradigm itself, rather than being an artifact of a specific backbone. Given its consistent top performance, we select MICD as the default feature embedding module for all subsequent experiments.

F. t-SNE Visualization of the CR Process

To intuitively validate the effectiveness of our proposed CR paradigm, we conduct a series of t-SNE visualization analyses. To this end, we visualize the relational representations derived from the generalized correlation matrix. For any given query sample, its relational representation is defined as the D-dimensional query–class vector from the matrix that corresponds to its ground-truth class label.

First, to demonstrate the overall impact of the RCRG blocks, we compare the relational structures before and after the refinement process. As shown in Fig. 11, the relational

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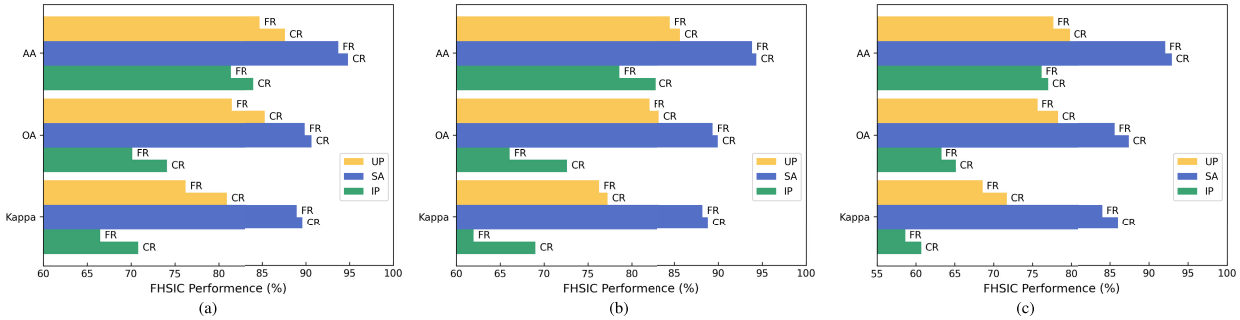


Fig. 10. Performance comparison between the *feature refinement (FR)* and *CR* paradigms across three HSI datasets and three feature embedding backbones. Each subplot presents results on the UP, SA, and IP datasets using a specific feature embedding backbone: (a) MICD [12], (b) CMFSL [8], and (c) Fast3DCNN [20]. Note that, here, *FR also refers to the RN-based method*. This demonstrates the superiority of our CR paradigm and the scalability and robustness of our PCRNet. Note also that we employ MICD as the default feature embedding module of our approach.

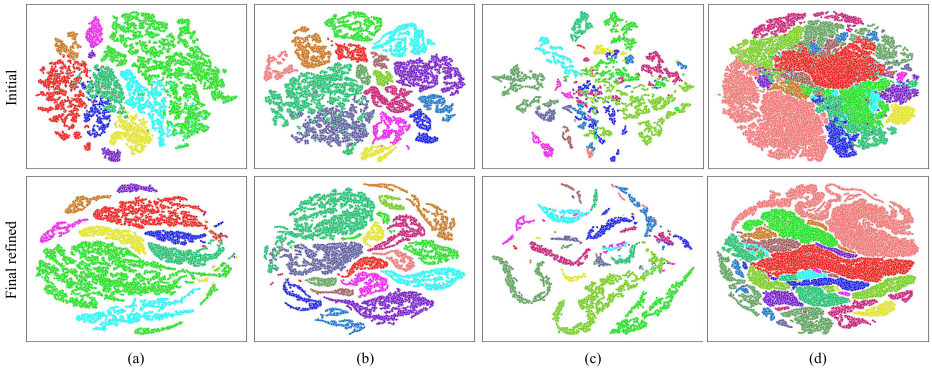


Fig. 11. t-SNE visualization comparing relational representations from the initial (top row) and final refined (bottom row) correlation matrices across four TD datasets. The refinement process transforms the initially entangled relational space into one with compact and clearly separated classes (a) The UP dataset. (b) The SA dataset. (c) The IP dataset. (d) The HC dataset.

representations extracted from the initial correlation matrix are entangled, with obvious overlap between classes, indicating a confusing relational space. In contrast, after being processed by three RCRG blocks, the representations from the final refined matrix form highly compact, well-separated intraclass clusters. This visual evidence confirms that our method effectively learns a well-structured correlation space, where relevant query–class pairs are pulled closer, while irrelevant ones are pushed apart.

Next, to reveal the progressive refinement nature of our PCRNet, we visualize the evolution of the correlation structure as it passes through the cascaded RCRG blocks. For clarity, we randomly select one class from each TD dataset and track its relational representations. In Fig. 12, green dots denote the vectors corresponding to the correct query–class pairs for this class, while gray dots represent all incorrect pairs. We can see that the green dots become increasingly concentrated with each additional RCRG block. This visualization explicitly illustrates the coarse-to-fine refinement mechanism at the heart of our model, confirming that our PCRNet progressively and effectively optimizes the correlation structure to enhance class discriminability.

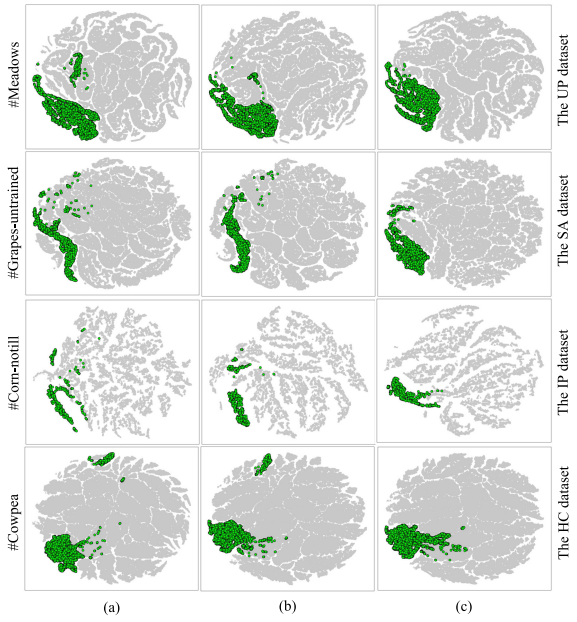


Fig. 12. Visualization of the progressive refinement of the correlation structure for a single class. Green dots represent correct query–class relations. The stages are labeled as "via RCRG 1" (after one block), "via RCRG 1&2" (after two blocks), and "via RCRG 1&2&3" (after three blocks), from left to right. As the number of RCRG blocks increases, the correct relations become more compact, demonstrating the iterative optimization capability of PCRNet. (a) Via RCRG 1. (b) Via RCRG 1&2. (c) Via RCRG 1&2&3.

G. Analysis of Computational Complexity

In this section, we evaluate the computational complexity of our proposed PCRNet in terms of model size, floating-point operations (FLOPs), and inference speed. As detailed in Table VII, our network remains lightweight, with only 0.98M parameters and 33.74M FLOPs, which ranks favorably against other state-of-the-art methods, underscoring the efficiency of

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TABLE VII
COMPARISONS OF NETWORK PARAMETERS (MILLION), FLOPS (MILLION), AND INFERENCE TIME (SECOND) ON THE TD DATASETS

Methods		Gia-CFSL	DA-CAA	CTF-SSCL	FDFSL	TEFSL	Ours
Parameters		0.31	1.01	0.40	0.33	0.95	0.98
FLOPs		198.86	67.92	6.37	69.84	57.24	33.74
Runtime	UP	12.10	20.87	3.82	9.28	16.94	11.13
	SA	114.27	26.88	6.98	15.82	32.80	22.71
	IP	3.52	5.08	1.67	2.03	7.86	4.26
	HC	34.28	244.92	35.98	230.85	277.12	103.32

our CR paradigm. This efficiency is also reflected in its fast inference speeds across all four benchmark datasets. Consequently, our PCRNet achieves a favorable balance between its superior classification accuracy and modest computational overhead, making it a practical and effective solution for FHSIC tasks.

V. DISCUSSION AND LIMITATION

To further elucidate the core novelty of our work, this section provides a comparative analysis distinguishing our proposed PCRNet from GNN and Transformer-based methods, and especially conventional RNs [42]. While GNN-based FHSIC methods [22], [23] typically model relations implicitly via feature propagation on a graph (where nodes represent samples), our PCRNet elevates the generalized correlation matrix itself to be the primary, explicit object of learning. As for Transformer-based methods [10], [11], [29], [62], although we adopt the attention architecture, our application represents a paradigm shift. Unlike Transformer-based HSIC methods that apply attention to image tokens for feature extraction, PCRNet applies attention to the correlation space (i.e., query–class relationships). This facilitates explicit relational reasoning rather than mere spatial–spectral feature enhancement, offering a novel perspective on utilizing attention in FSL.

While both PCRNet and RN-based methods aim to learn the interaction relationships between query and support samples, their underlying philosophies diverge on three critical levels: the modeling paradigm, the operational object, and the capability boundary. Conventional RN-based FHSIC methods [43], [44], [45], [46] typically operate within the "representation-metric" framework. It advocated learning a superior, nonlinear distance metric function through a learnable deep network (also known as a relation module). The operational object is an isolated pair of concatenated feature vectors, which is processed to directly regress a scalar relation score. This constitutes a context-agnostic, one-step mapping from features to a final score. In contrast, our PCRNet establishes a "relation-reasoning" framework. Its primary goal is not to measure features, but to iteratively refine a holistic relational structure. The operational object is the entire $N_q \times N_c \times D$ generalized correlation matrix, which represents the complete state of query–class relationships. Instead of a single mapping, PCRNet employs cascaded RCRG blocks to progressively enhance this matrix. Dedicated modules such as QAM and CGA dynamically modulate these correlations by incorporating diverse contextual information, such as interquery dependencies and dataset-wide global class knowledge—capabilities that are structurally absent in the one-pass, isolated computation of the standard relation module of RN-based methods.

Particularly, a key contribution of our work is the explicit proposal of CR as a distinct FHSIC paradigm. Unlike RN-based methods, which merely seek a learnable substitute for distance metrics, our paradigm treats correlation itself as a core, structured state to be iteratively optimized. This *perspective shift* opens up new avenues for model design, moving beyond feature enhancement toward explicit relational reasoning. This conceptual advancement is empirically validated by the significant performance gains of PCRNet over RN-based FHSIC methods, as shown in Fig. 10.

Despite the superior classification performance achieved via this paradigm shift, a limitation remains in PCRNet regarding high interclass spectral similarity. In certain scenarios, distinct classes may exhibit extremely similar spectral signatures, rendering the initial correlations highly ambiguous. In such cases, this excessive ambiguity may exceed the rectification capabilities of our "relation-reasoning" framework, posing a challenge to the accurate discrimination of subtle spectral discrepancies. More analysis is provided in Section III of the supplementary material. In future work, we intend to explore the applicability and potential of our CR paradigm across a broader range of FSL tasks. In addition, we aim to investigate more robust initialization strategies and adaptive refinement mechanisms to better handle these complex scenarios.

VI. CONCLUSION

In this article, we have proposed a novel CR paradigm to advance the field of FHSIC. Our proposed PCRNet embodies this paradigm by shifting the learning focus from conventional feature refinement to the explicit modeling and refinement of structured relationships between support and query samples. Specifically, PCRNet begins by constructing a learnable generalized correlation matrix to encode its interactions. Subsequently, this matrix is iteratively enhanced via a synergistic process: On the one hand, MGA injects class commonalities with cross-domain robustness from a holistic perspective; on the other hand, QAM focuses on capturing fine-grained dependencies among query samples. These complementary insights are then adaptively fused by SGI. Through this coarse-to-fine refinement, the network learns a more discriminative correlation representation to guide more robust classification, a capability absent in previous relation-based methods. We believe that this perspective shift opens a new avenue for developing more context-aware and robust models.

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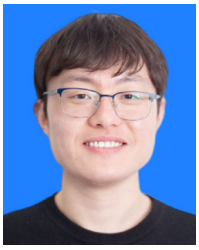
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